Introduction to Artificial Intelligence

Prof. Dr. Hamed Shariat Yazdi, Darmstadt University of Applied Sciences

Assignment:
Machine learning – Regression

In this lab session, you will practice ML regression with the Kaggle challenge "TMDB box office prediction" https://www.kaggle.com/c/tmdb-box-office-prediction . You will be working in your teams of two students each.

1. If not yet done, read Sections 2.1-2.4 of the book "Applied Artificial Intelligence – An Engineering Approach". Focus on the paragraphs relevant for this course.

2. Prepare the quick check questions

3. Download the Kaggle training data, test data and sample submission from https://www.kaggle.com/c/tmdb-box-office-prediction/data

4. Analyze the dataset.

5. In the Pycharm project for your team, add a new Python package "03b_ML_regression_TMDB" and add a new Juypter notebook "TMDB.ipynb in this package

6. Copy the data files to a subfolder "data"

7. Implement the Kaggle challenge

8. Perform a cross-validation on the training data. You may improve your score by modifying your implementation, e.g., pre-processing, other ML approaches, other ML parameters etc.

9. Submit the submission file generated by your implementation to Kaggle

10. Check your position in the leaderboard.

11. Commit your code to the Git repository

12. Add your team number, Kaggle score (float) and Kaggle position (int) in the shared spreadsheet (link in Moodle), tab "03b ML regression leaderboard"